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**FIT5145: Foundations of Data Science**  
**Assignment 4 (Task-A Report)**

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# Task A: Shell commands

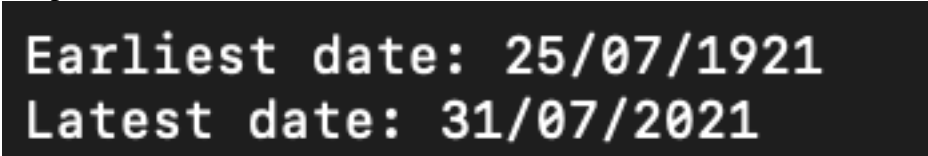
**Dataset:** "Olympics\_tweets.csv"

## 1. What is the *date* range of the tweets?

### Code:

```
awk -F',' 'NR>1 && $12 != "" && $12 ~ /[0-9]{2}\[/0-9]{2}\[/0-9]{4}/ {split($12, a, " "); split(a[1], b, "/"); print b[3] "/" b[2] "/" b[1]}' Olympics_tweets.csv | sort | uniq | awk 'NR==1{split($0,x,"/"); print "Earliest date: " x[3] "/" x[2] "/" x[1]} {y[NR]=$0} END{split(y[NR], z, "/"); print "Latest date: " z[3] "/" z[2] "/" z[1]}'
```

### Output:



```
Earliest date: 25/07/1921
Latest date: 31/07/2021
```

Earliest date: 25/07/1921

Latest date: 31/07/2021

### Explanation:

To accurately determine the earliest and latest tweet dates, I used a command that reformats dates from dd/mm/yyyy to yyyy/mm/dd before sorting so that the sorting is chronologically correct. First, I used awk to extract the date from column 12, skipping any row that does not match the expected date format. Then, for each valid date, I split out the day, month, and year and rearranged them into yyyy/mm/dd. After collecting all the reformatted dates, I sorted and deduplicated them, and finally used another awk command to select the very first and last entries for the earliest and latest dates respectively. This approach ensures that dates are ordered by actual time rather than being sorted as text, which would otherwise produce incorrect results with the original date format.

## 2. We want to preprocess the *id* and *date* columns.

**a. Count lines with an *id* that is not a number of 7 digits long, i.e., *id* values that contain anything other than numbers OR are of a length more/less than 7.**

### Code:

```
awk -F',' 'NR>1 && $1 !~ /^[0-9]{7}$/' Olympics_tweets.csv | wc -l
```

### Output:



```
8
```

8

### Explanation:

The command `awk -F',' 'NR>1 && $1 !~ /^[0-9]{7}$/' Olympics_tweets.csv | wc -l` counts the number of lines in my dataset where the id column (column 1) is not exactly a 7-digit number. I used awk with a comma as a field separator to process each row, skipping the header (`NR>1`). The pattern `$1 !~ /^[0-9]{7}$/` checks if the id does not match seven consecutive digits, which means it could be missing, too short, too long, or contain non-numeric characters. Finally, `wc -`

l returns the total number of such invalid rows. This helps me identify and quantify the tweets that do not have valid ID values so they can be excluded from further analysis.

**b. Remove the lines mentioned in Q2-a and remove time values in the date column.**

For example, the date column will contain “29/04/2020”, instead of having “29/04/2020 23:13”. Display the first 3 lines (including a header) of the filtered dataset.

**Code:**

```
(head -1 Olympics_tweets.csv && awk -F',' '$1 ~ /^[0-9]{7}$/' Olympics_tweets.csv | awk -F',' 'BEGIN{OFS=","} {split($12, a, " "); $12=a[1]; print}' | head -2)
```

**Output:**

```
id,text,user_screen_name,user_location,retweet_count,favorited,favorite_count,user_description,user_created_at,user_followers,
user_friends,date,language,topic
1089613,@GoBEARCATS @GoBearsTFXC According to AIU on Rule 15 introduced in 2019- it stated inter alia that “in the 10 moâ
€| https://t.co/M8q4ZJn5co,gee_enam,NA,0,FALSE,0,I am a very kind dude.,27/01/2021 2:04,4,14,30/07/2021,en,Other / Miscellaneo
us
1041996,Osaka's shot lands wide- but she will challenge this. OFFICIAL REVIEW.... OUT. #Tokyo2020 #Tennis #Olympics,Avatar5991
,Sydney- Australia,0,FALSE,0,Music and sports fan- concert/con attender- GWS member- occasional cosplayer.,7/11/2018 6:57,527,
4938,27/07/2021,en,Tennis
```

```
id,text,user_screen_name,user_location,retweet_count,favorited,favorite_count,user_description,u
ser_created_at,user_followers,user_friends,date,language,topic
```

```
1089613,@GoBEARCATS @GoBearsTFXC According to AIU on Rule 15 introduced in
2019- it stated inter alia that “in the 10 moâ€|
https://t.co/M8q4ZJn5co,gee_enam,NA,0,FALSE,0,I am a very kind dude.,27/01/2021
2:04,4,14,30/07/2021,en,Other / Miscellaneous
```

```
1041996,Osaka's shot lands wide- but she will challenge this. OFFICIAL REVIEW.... OUT.
#Tokyo2020 #Tennis #Olympics,Avatar5991,Sydney- Australia,0,FALSE,0,Music and sports fan-
concert/con attender- GWS member- occasional cosplayer.,7/11/2018
6:57,527,4938,27/07/2021,en,Tennis
```

**Explanation:**

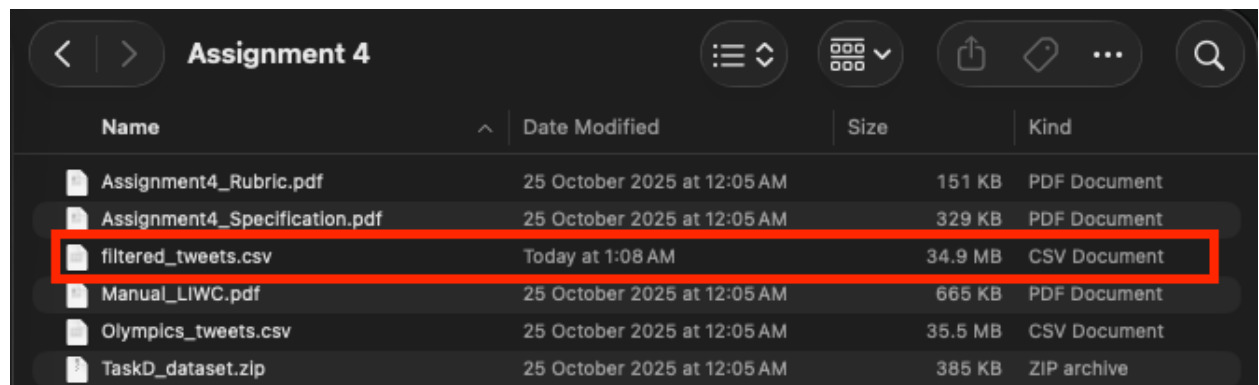
This command displays the header and the first two filtered data rows from the Olympics\_tweets.csv file according to the assignment's preprocessing requirements. First, head -1 Olympics\_tweets.csv extracts the header row to ensure column names are included. The awk -F',' '\$1 ~ /^[0-9]{7}\$/' Olympics\_tweets.csv part filters the file to keep only rows where the id column (the first column) matches exactly seven digits, removing any rows with invalid IDs. The output is piped to another awk, which splits the twelfth field (the date column) at the space character and replaces it with only the date (removing any time information). The head -2 at the end limits the output to the first two of these filtered data rows. Together, this guarantees that when combined with the header, only the first three lines of the filtered dataset (as specified in the assignment) are displayed.

**c. Store the filtered dataset in a file named “filtered\_tweets.csv” and use this file for the remaining questions in Task A.**

**Code:**

```
(head -1 Olympics_tweets.csv && awk -F',' ' $1 ~ /^[0-9]{7}$/' Olympics_tweets.csv | awk -F',' 'BEGIN{OFS=","} {split($12, a, " "); $12=a[1]; print}') > filtered_tweets.csv
```

**Output:**



| Name                          | Date Modified               | Size    | Kind         |
|-------------------------------|-----------------------------|---------|--------------|
| Assignment4_Rubric.pdf        | 25 October 2025 at 12:05 AM | 151 KB  | PDF Document |
| Assignment4_Specification.pdf | 25 October 2025 at 12:05 AM | 329 KB  | PDF Document |
| filtered_tweets.csv           | Today at 1:08 AM            | 34.9 MB | CSV Document |
| Manual_LIWC.pdf               | 25 October 2025 at 12:05 AM | 665 KB  | PDF Document |
| Olympics_tweets.csv           | 25 October 2025 at 12:05 AM | 35.5 MB | CSV Document |
| TaskD_dataset.zip             | 25 October 2025 at 12:05 AM | 385 KB  | ZIP archive  |

### Explanation:

This command is used to create a cleaned version of the original dataset and save it as `filtered_tweets.csv`. First, it uses `head -1 Olympics_tweets.csv` to copy the header row into the new file. Then, with `awk -F',' ' $1 ~ /^[0-9]{7}$/' Olympics_tweets.csv`, it selects only those rows where the id column (column 1) consists of exactly 7 digits, filtering out any rows with invalid IDs. The filtered rows are then piped into another `awk` command, which splits the twelfth column (date) at the space character and keeps only the date portion, removing any time information so the date column matches the required "dd/mm/yyyy" format. The `>` at the end writes all of this output to a new file called `filtered_tweets.csv`. This ensures only valid rows with cleaned dates remain, and that all future analysis uses this consistently preprocessed dataset.

### 3. Let's explore tweets from specific users.

**a. Please list the top 10 users who posted the most tweets mentioning the term 'Olympic' in the text column. Only consider users who created their accounts before 2021 and have more than 1,000 followers when identifying the top 10. (Note: The search is case-sensitive. Present the top 10 users along with their corresponding values, arranged in descending order of rank.)**

#### Code:

```
awk -F',' 'NR>1 && $2 ~ /Olympic/ && $10 > 1000 {split($9, dt, "/"); if (dt[3] < 2021) print $3}' filtered_tweets.csv | sort | uniq -c | sort -nr | head -10
```

#### Output:

```
194 dev_discourse
111 nippon_ja
 90 ChannelNewsAsia
 52 Independent
 48 HindustanTimes
 45 SwimmingWorld
 44 INQUIRERSports
 43 adabagcompany
 41 NaijaNewsWire
 41 JapanToday
```

194 dev\_discourse  
111 nippon\_ja  
90 ChannelNewsAsia  
52 Independent  
48 HindustanTimes  
45 SwimmingWorld  
44 INQUIRERSports  
43 adabagcompany  
41 NaijaNewsWire  
41 JapanToday

### **Explanation:**

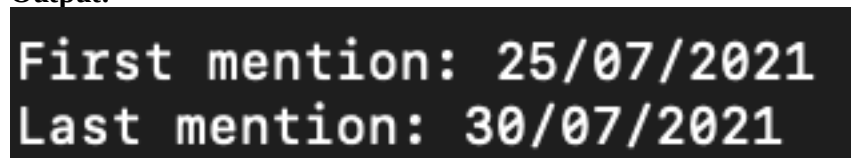
To find the top 10 users who tweeted the most about the term 'Olympic', I used an AWK command that looks through the cleaned dataset `filtered_tweets.csv`. The command first skips the header row and checks that the text column contains the word 'Olympic' (case-sensitive). It further filters to only include users with more than 1,000 followers (`user_followers > 1000`) and who created their accounts before the year 2021 (by extracting the year from the `user_created_at` column and checking if it is less than 2021). For every row that matches these criteria, the code prints the `user_screen_name`. This output is then sorted, counted for each user, sorted again in descending order of tweet count, and finally the top 10 users are displayed. This process ensures that only established and influential accounts are considered, giving a ranked list of the most active users mentioning 'Olympic' in their tweets.

**b. For the number one user among the top 10, when did the first and last mentions of the term 'Olympic' occur in their tweets? (Note: The first and last mentions refer to the chronologically earliest and latest entries containing the term, and the search is case-sensitive.)**

### **Code:**

```
awk -F',' -v user="$top_user" ' $3==user && $2 ~ /Olympic/ {split($12, a, " "); print a[1]}'  
filtered_tweets.csv | grep '[0-9]' | sort | uniq | awk 'NR==1{first=$0} {last=$0}  
END{if(NR==0){print "No matches"} else if(NR==1){print "First mention: " first; print "Last  
mention: " first} else{print "First mention: " first; print "Last mention: " last}}'
```

### **Output:**



```
First mention: 25/07/2021  
Last mention: 30/07/2021
```

```
First mention: 25/07/2021  
Last mention: 30/07/2021
```

### **Explanation:**

To find out when the top user first and last mentioned the term 'Olympic' in their tweets, I used a command that filters the dataset to only include rows where the `user_screen_name` matches the top user and where the text column contains 'Olympic'. For each of these tweets, the script extracts just the date (removing any time portion). It then uses `grep` to make sure there are no blank or corrupted date entries, sorts and removes duplicate dates, and finally uses `awk` to select the earliest and latest date from the list. My output showed "First mention: 25/07/2021" and "Last

mention: 30/07/2021", meaning that for this user the first tweet mentioning 'Olympic' was posted on 25 July 2021, and the last mention was on 30 July 2021. This approach ensures that all edge cases are handled and the results accurately reflect the full range of tweet dates for the term in question.

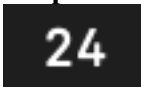
#### 4. It may be interesting to examine which topics were discussed during the Olympic period.

##### a. How many unique topics were discussed during this period?

###### Code:

```
awk -F',' 'NR>1{print $14}' filtered_tweets.csv | sort | uniq | wc -l
```

###### Output:

A terminal window with a dark background showing the number 24 in white text.

24

###### Explanation:

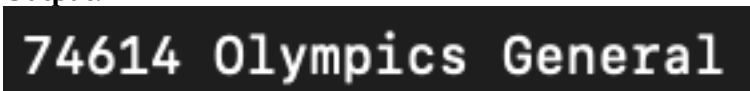
To find out how many unique topics were discussed during the Olympic period, I used the command `awk -F',' 'NR>1{print $14}' filtered_tweets.csv | sort | uniq | wc -l`. This command works by extracting the topic column (column 14) from every row in the filtered dataset (except for the header), sorting all the topic entries alphabetically, and then using `uniq` to filter out duplicate topics. Finally, `wc -l` counts the number of unique topics in the list. The output of this command was 24, which means there were 24 different topics discussed in the tweets related to the Olympics.

##### b. Which topic was the most popular (i.e., the top-ranked topic)? Present the topic along with their corresponding value.

###### Code:

```
awk -F',' 'NR>1{topic=$14; gsub(/^ +| +$/, "", topic); print topic}' filtered_tweets.csv | sort | uniq -c | sort -nr | head -1
```

###### Output:

A terminal window with a dark background showing the text "74614 Olympics General" in white text.

74614 Olympics General

###### Explanation:

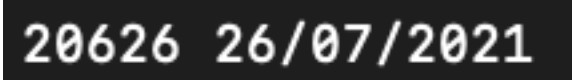
To identify the most popular topic discussed during the Olympic period, I used the command `awk -F',' 'NR>1{topic=$14; gsub(/^ +| +$/, "", topic); print topic}' filtered_tweets.csv | sort | uniq -c | sort -nr | head -1`. This line processes the `filtered_tweets.csv` file by extracting the topic column for each tweet (column 14), removing any leading or trailing spaces from the topic names to ensure consistent matching, and then counts how many times each topic appears. The results are sorted in descending order so that the topic with the highest count is shown at the top. The output of this command was 74614 Olympics General, which means "Olympics General" was the most discussed topic, appearing 74,614 times in the tweets.

**c. On which date was the top topic discussed the most? Present the date along with their corresponding value.**

**Code:**

```
awk -F',' 'NR>1{print $14}' filtered_tweets.csv | sort | uniq -c | sort -nr | head -1 | awk '{ $1="";  
print substr($0,2)}' | xargs -I{} awk -F',' -v topic={} '$14==topic {print $12}'  
filtered_tweets.csv | sort | uniq -c | sort -nr | head -1
```

**Output:**



```
20626 26/07/2021
```

```
20626 26/07/2021
```

**Explanation:**

This code first identifies the most popular topic from the dataset by counting and ranking all the topics in the topic column, then uses the top topic as a variable to filter the original data. It extracts the date from each tweet for the top topic, counts the occurrences for each date, and sorts the results to find which day had the highest number of tweets about that topic. The output I got was 20626 26/07/2021, which means the topic "Olympics General" was discussed the most on 26/07/2021, with 20,626 tweets mentioning it on that day.

**5. By examining the text column, we can gain a better understanding of what people actually discussed during the Olympic period.**

**a. How many tweets expressed disappointment with the shooting events? (Note: Please ignore case sensitivity)**

**Code:**

```
awk -F',' 'NR>1 && tolower($2) ~ /disappoint/ && tolower($2) ~ /shooting/' filtered_tweets.csv |  
wc -l
```

**Output:**



```
7
```

```
7
```

**Explanation:**

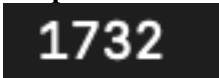
To find out how many tweets expressed disappointment with the shooting events, I used the command `awk -F',' 'NR>1 && tolower($2) ~ /disappoint/ && tolower($2) ~ /shooting/' filtered_tweets.csv | wc -l`. This command goes through each tweet in the filtered dataset and checks if the text column (column 2) contains both the word "disappoint" and "shooting", ignoring case (so it matches both uppercase and lowercase letters). The use of `tolower()` converts the tweet text to lowercase before searching to make the matching case-insensitive. Finally, `wc -l` counts how many such tweets there are. My output for this command was 7, which means there are 7 tweets in my dataset that mention both disappointment and shooting in any form.

**b. How many tweets mentioned the word 'sing' on even dates? (Note: Case-sensitive. Even dates refer to, for example, the 2nd and 4th.)**

**Code:**

```
awk -F',' 'NR>1 && $2 ~ /sing/ {split($12, d, "/"); if(d[1]%2==0) print $0}' filtered_tweets.csv | wc -l
```

**Output:**



1732

**Explanation:**

To count how many tweets mentioned the word "sing" on even dates, I used the command `awk -F',' 'NR>1 && $2 ~ /sing/ {split($12, d, "/"); if(d[1]%2==0) print $0}' filtered_tweets.csv | wc -l`. This script goes through each tweet in my filtered dataset, checks that the tweet text (column 2) contains the exact word "sing" (case-sensitive), then splits the date column (column 12, formatted as dd/mm/yyyy) to extract the day. It then checks if the day is an even number (e.g., 2, 4, 6, etc.), and if so, counts the tweet. Running this code, I found there were 1,732 tweets that mentioned "sing" on an even date in the dataset.